



## ANNEXURE A

# Complete Security Scenario Register

Power-generation surveillance, security and control-room response scenarios. All 26 records retain the original zone, applicability, detection logic, operational importance and response sequence.

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Generation and electrical assets

4

Material and movement integrity

7

Safety and process events

6

Remote and distributed assets

4

System integrity and investigation

# One consistent operating logic

01

## Detect

Field devices identify the condition or event.

02

## Correlate

The ICCC adds plant, identity, permit and process context.

03

## Respond

Operators execute and document the verified response.

# Generation and electrical assets

01

📍 ZONE A — PERIMETER

## Perimeter intrusion

Thermal / PTZ · line crossing → PIDS + GIS map → Guard dispatch

### WHAT IT DETECTS

Fence climb, cut or crossing of the outer boundary line detected by thermal cameras

### WHY IT MATTERS

The boundary is the first and cheapest place to stop an event; late detection converts a

02

📍 ZONE B — GENERATION AND ELECTRICAL

## Intrusion near switchyard / transformer zone

Fixed / PTZ · intrusion → GIS + work permit → Security alert

### WHAT IT DETECTS

Human presence inside the switchyard fence line, transformer bays or cable galleries outside sanctioned work windows

### WHY IT MATTERS

A single transformer or bay compromise can force a unit trip or partial generation loss, with long replacement lead times

03

📍 ZONE B — FUEL HANDLING

THERMAL

## Unauthorised approach to coal conveyor or CHP

Fixed · person detection → CHP clearance + ACS → CHP control room

### WHAT IT DETECTS

Approach to conveyor galleries, transfer points, crusher house or junction towers by

### WHY IT MATTERS

CHP areas combine nip-point fatality risk with fuel-pilferage exposure on a

04

📍 ZONE B — GENERATION

## Contractor entering restricted generation block

Dome · person / face → ACS + contractor record → Engineer-in-charge

### WHAT IT DETECTS

Contractor badge or face seen in turbine hall, boiler area or unit block outside the sanctioned scope, area or time window

### WHY IT MATTERS

Contractor movement discipline is the most frequent breakdown point in plant access control and audit findings

05

📍 ZONE C — CONTROL AND IT

## Unauthorised access to control-room approaches

Dome · person detection → ACS interlock + OT SOC → Shift charge engineer

### WHAT IT DETECTS

Movement in unit control room lobbies, UPS and relay rooms, or server and SCADA room approaches without dual authorisation

### WHY IT MATTERS

Physical proximity to control and OT assets is a direct sabotage and cyber-physical exposure path

# Material and movement integrity

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📍 ZONE B — CABLE TRENCHES AND YARDS

## Cable and copper theft

Fixed / thermal · intrusion → PIDS + zone map → Patrol

### WHAT IT DETECTS

Night-time presence, digging or trench-cover disturbance along cable routes, earth-mat areas and idle equipment yards.

### WHY IT MATTERS

Copper and cable loss carries a disproportionate restoration cost and can disable protection or auxiliary supply.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Thermal alert on trench route
- 2 Confirm no permitted work
- 3 Dispatch patrol to segment
- 4 FIR-grade evidence pack

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📍 ZONE A — GATES AND INTERNAL ROADS

## Wrong vehicle entering fuel or hazardous-material route

ANPR / LPR · vehicle → ACS + weighbridge → Gate denial

### WHAT IT DETECTS

ANPR mismatch, unapproved lane use or a vehicle diverging from the declared fuel, oil, chemical or ash route.

### WHY IT MATTERS

Route deviation drives both fuel quantity-integrity loss and hazardous-material safety exposure inside the plant.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 ANPR flag at gate
- 2 Stop and verify challan
- 3 Weighbridge cross-check
- 4 Route deviation report

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📍 ZONE C — STORES AND SCRAP

## Material removal from stores / scrap area

Fixed · object / loading → Gate pass + ERP → Stores security

### WHAT IT DETECTS

Loading, hand-carry or after-hours movement of material from stores yards, scrap bays or dismantled-equipment areas.

### WHY IT MATTERS

Scrap and spares movement is where value leakage is hardest to reconstruct after the fact without indexed video.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Alert stores security post
- 2 Match against gate pass
- 3 Retrieve indexed video clip
- 4 Raise deviation to materials head

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📍 ZONE A — APPROACHES AND GATES

## Suspicious vehicle behaviour

ANPR / PTZ · behaviour → ACS + watchlist → Verification

### WHAT IT DETECTS

Repeat passes, prolonged dwell near boundary or gates, or a plate appearing on a watchlist across multiple approach cameras.

### WHY IT MATTERS

Pre-attack reconnaissance and organised pilferage both show up as vehicle behaviour long before an intrusion.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Behaviour alert to ICC
- 2 Watchlist and history check
- 3 Field verification by patrol
- 4 Escalate to district police if repeated

# Safety and process events

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📍 ZONE B — TRANSFORMERS AND OIL STORAGE

## Transformer / oil-area fire and smoke corroboration

Thermal · heat / flame → FAS + SCADA → Emergency response

WHAT IT DETECTS

WHY IT MATTERS

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📍 ZONE B — MILLS, CONVEYORS, CABLE GALLERIES THERMAL

## Thermal hotspot on rotating or conveying equipment

Thermal · anomaly → SCADA / FAS → Inspection

WHAT IT DETECTS

WHY IT MATTERS

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📍 ZONE B — PLANT AREAS AND STRUCTURES

## Person-down / worker fall

Fixed · person-down → Safety + occupancy → Emergency medical

WHAT IT DETECTS

WHY IT MATTERS

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📍 ZONE B — MANDATED PPE AREAS

## PPE violation in mandated zone

Fixed · PPE analytic → HSE + contractor record → Supervisor

WHAT IT DETECTS

WHY IT MATTERS

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📍 ZONE B — HAZARDOUS AREAS

## Personnel remaining inside hazardous zone after shutdown

Fixed · occupancy → Permit + LOTO → Re-energisation block

WHAT IT DETECTS

WHY IT MATTERS

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📍 ZONE B — EX-RATED AREAS THERMAL / GAS

## Gas-area intrusion in Ex-rated zone

Ex-rated camera · intrusion → Gas detection → Emergency protocol

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📍 ZONE B — BATTERY ENERGY STORAGE

SOLAR / BESS (WHERE APPLICABLE)

## BESS thermal event

Thermal · heat anomaly → BMS / FAS → Emergency protocol

# Remote and distributed assets

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📍 ZONE B — PV ARRAY FIELD **SOLAR**

## Solar-field intrusion and module theft

Thermal / PTZ · intrusion → GIS array map → Patrol

### WHAT IT DETECTS

Boundary breach or movement between panel rows, inverter stations and module

### WHY IT MATTERS

Module and cable theft across a sparsely manned field is a steady generation and

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📍 ZONE C — WATER SYSTEMS **HYDRO / THERMAL**

## Dam, intake and reservoir approach

Long-range PTZ · intrusion → GIS + intake control → Patrol

### WHAT IT DETECTS

Boats, swimmers, unauthorised persons or

### WHY IT MATTERS

Cooling-water or hydro intake interference

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📍 ZONE A — AIRSPACE

## Drone approach towards critical assets

EO / IR + RF · UAV track → Counter-UAS → Designated agency

### WHAT IT DETECTS

RF or radar detection of a UAV track approaching switchyard, cooling towers, boiler structures or fuel storage

### WHY IT MATTERS

Aerial reconnaissance and payload risk over critical generation assets is a declared national-security concern for sensitive

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📍 ZONE B — REMOTE ENCLOSURES

**WIND / REMOTE (WHERE APPLICABLE)**

## Remote asset / turbine enclosure access

Fixed · door / access → ACS + work order → Alert and verify

### WHAT IT DETECTS

Door opening or presence inside unmanned

### WHY IT MATTERS

Unmanned assets are attacked precisely

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📍 ZONE B — WATER-CONDUCTOR SYSTEM **HYDRO**

## Tunnel and water-conductor system intrusion

Thermal / fixed · intrusion → Access control → Controlled response

### WHAT IT DETECTS

Entry at adits, surge shafts, penstock galleries or tunnel portals outside inspection

### WHY IT MATTERS

Tunnel access is both a sabotage route to the machine hall and a confined-space

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📍 ZONE A — GATES

## Crowd build-up at gate during labour disturbance

PTZ · crowd density → PSOC + HR / IR → Security and admin

### WHAT IT DETECTS

Crowd density and dwell-time thresholds crossed at main gate, contractor gate or administrative approaches

### WHY IT MATTERS

Gate obstruction during an industrial-relations event is a direct threat to coal receipt, shift change and generation

# System integrity and investigation

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📍 ZONE A — PERIMETER AND GATES

## Camera loss or tampering at critical coverage

Fixed · tamper / signal loss → VMS health → Maintenance SLA

### WHAT IT DETECTS

Signal loss, defocus, spray, obstruction or reorientation on perimeter, switchyard and gate cameras.

### WHY IT MATTERS

Deliberate blinding of critical coverage is usually a precursor event, and silent camera loss defeats the whole estate.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Health alarm with criticality
- 2 Adjacent-camera
- 3 Technician ticket with SLA
- 4 Uptime reporting to

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📍 ZONE A AND C — GATES, ADMIN, PUBLIC INTERFACE

## Abandoned object or unattended package

Fixed / PTZ · object left → PSOC review → Investigation

### WHAT IT DETECTS

Bag, package or unattended equipment remaining static beyond threshold near gates, admin blocks or transport bays.

### WHY IT MATTERS

Unattended-object protocol is a baseline expectation for critical infrastructure and audit readiness.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Object alert with dwell time
- 2 Backtrack who placed it
- 3 Cordon and inform security head
- 4 Bomb-disposal protocol if unresolved

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📍 ZONE B — PLANT EQUIPMENT

## Equipment tampering on critical assets

Fixed · object / tamper → CMMS + ACS → Maintenance and security

### WHAT IT DETECTS

Interference with panels, valves, relay enclosures, earth connections or metering and instrumentation.

### WHY IT MATTERS

Tampering blends into maintenance activity and is only separable when video is indexed against work orders.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Tamper alert with asset tag
- 2 Match against CMMS work order
- 3 Joint maintenance and security
- 4 Root-cause and access tightening

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📍 PLANT-WIDE

## Multi-camera post-event investigation

Multiple · tracking / search → VMS + AI search → Investigation pack

### WHAT IT DETECTS

Appearance and attribute search to reconstruct a person or vehicle track across the full camera estate and time window.

### WHY IT MATTERS

Investigation speed determines whether an incident becomes an accountable case or an unexplained loss in the register.

### 🛡️ CONTROL-ROOM RESPONSE

- 1 Define time and area window
- 2 Attribute / appearance
- 3 Assemble chain-of-
- 4 Close case in incident register